

Lattice gauge theory notes

Steven Cotterill
(Dated: January 2026)

I. THEORY

A. Global case

It is convenient in this case to derive the equations of motion in the continuum limit and then discretise equations of motion. Starting from a simple field theory with no local symmetries,

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi^a \partial^\mu \phi^a - V(\phi^a), \quad (1.1)$$

the equations of motion can be derived to be

$$\partial_\mu \partial^\mu \phi^a + \frac{\partial V}{\partial \phi^a} = 0. \quad (1.2)$$

From here, the only remaining requirement in order to be able to simulate the system on a lattice is to define a finite difference operator to replace $\partial_\mu \partial^\mu$. For example, one could use a 3-point stencil in a cartesian coordinate system so that

$$\partial_\mu^2 \phi^a \rightarrow \frac{\phi^a(\mathbf{x} + \mathbf{s}_\mu) - 2\phi^a(\mathbf{x}) + \phi^a(\mathbf{x} - \mathbf{s}_\mu)}{s_\mu^2}, \quad (1.3)$$

where we have defined $\mathbf{s}_\mu = s_\mu \hat{\mathbf{s}}_\mu$ to be vectors pointed along the coordinate directions of the lattice, with length corresponding to the lattice spacing in each direction.

It is useful to discuss an alternate way of deriving these discretised equations of motion, which is equivalent in the global theory but we will later see is not equivalent in gauged theories. The key difference is to start by discretising the Lagrangian, after which we can still proceed by requiring that the action is extremised. Instead of defining a finite difference operator for the second order derivatives, in the Lagrangian we need the first order derivatives. We will choose to use a forward finite difference,

$$\partial_\mu \phi^a \rightarrow \frac{\phi^a(\mathbf{x} + \mathbf{s}_\mu) - \phi^a(\mathbf{x})}{s_\mu}, \quad (1.4)$$

as a simple example, but higher order finite differences can also be used. First, let us confirm that this does indeed approximate the derivative, which can also indicate how accurate the approximation is. By Taylor expanding the field at the neighbouring grid point, we have that $\phi^a(\mathbf{x} + \mathbf{s}_\mu) = \phi^a(\mathbf{x}) + s_\mu \partial_\mu \phi^a(\mathbf{x}) + \mathcal{O}(s^2)$. Substituting into our definition for the finite difference gives us

$$\frac{\phi^a(\mathbf{x} + \mathbf{s}_\mu) - \phi^a(\mathbf{x})}{s_\mu} = \partial_\mu \phi^a(\mathbf{x}) + \mathcal{O}(s), \quad (1.5)$$

so our finite difference does approximate the derivative, but with errors that are of the order of the lattice spacings. Nevertheless, we can now write down the discretised Lagrangian as

$$\mathcal{L} = \sum_{\mathbf{x}} \left\{ \frac{1}{2} \eta_{\mu\nu} \left(\frac{\phi^a(\mathbf{x} + \mathbf{s}_\mu) - \phi^a(\mathbf{x})}{s_\mu} \right) \left(\frac{\phi^a(\mathbf{x} + \mathbf{s}_\nu) - \phi^a(\mathbf{x})}{s_\nu} \right) - V(\phi^a(\mathbf{x})) \right\}, \quad (1.6)$$

where the sum is over all lattice sites and we have explicitly included the Minkowski metric, $g_{\mu\nu}$, so that we can make simplifications in the following analysis. We can now find the equations of motion by taking the derivative with respect to $\phi(\mathbf{x})$ and, after making sure to include the contributions from neighbouring sites, we get

$$\frac{\phi^a(\mathbf{x} + \mathbf{s}_0) - 2\phi^a(\mathbf{x}) + \phi^a(\mathbf{x} - \mathbf{s}_0)}{s_0^2} - \frac{\phi^a(\mathbf{x} + \mathbf{s}_i) - 2\phi^a(\mathbf{x}) + \phi^a(\mathbf{x} - \mathbf{s}_i)}{s_i^2} + \frac{\partial V}{\partial \phi^a(\mathbf{x})} = 0, \quad (1.7)$$

where I have taken $\mathbf{s}_0$ to point in the timelike direction and there is an implicit sum over i . This is the same equation of motion that we arrived at previously, but here the stencil for the second order derivative has naturally appeared based on our choice for the first order finite difference.

B. Gauged case

In the gauged case, the two methods discussed above do not give the same results and attempting to naively impose the continuum form of the equations of motion onto the lattice directly breaks the symmetry associated with the gauge transformation. In order to see why, it is useful to revisit some of the basic features of local symmetries.

Suppose that the potential for our model has a symmetry under some transformation $\Omega \in G$ acting on Φ and neglecting the indices for the internal degrees of freedom in Φ . Mathematically, this means that $V(\Omega\Phi) = V(\Phi)$. However, this transformation is only a symmetry of the Lagrangian if the transformation is global — Ω does not depend on position¹.

If we wish for local symmetry transformations, $\Omega(\mathbf{x})$, to be a symmetry of the Lagrangian, then we need to promote the derivatives to covariant derivatives. In the continuum limit, this can be done by replacing $\partial_\mu\Phi$ with

$$\mathcal{D}_\mu\Phi = \partial_\mu\Phi + iA_\mu\Phi, \quad (1.8)$$

such that $(\mathcal{D}_\mu\Phi)^\dagger(\mathcal{D}^\mu\Phi)$ is invariant under gauge transformations, $\Phi = \Omega\tilde{\Phi}$ and $A_\mu = \Omega A_\mu\Omega^\dagger + i(\partial_\mu\Omega)\Omega^\dagger$. However, if we try to directly implement this on the lattice, by replacing continuum derivative operators with finite differences, we will see that we will not retain the gauge symmetry,

$$\begin{aligned} \mathcal{D}_\mu\Phi &\rightarrow \frac{\Phi(\mathbf{x} + \mathbf{s}_\mu) - \Phi(\mathbf{x})}{s_\mu} + iA_\mu(\mathbf{x})\Phi(\mathbf{x}) \\ &= \frac{\Omega(\mathbf{x} + \mathbf{s}_\mu)\tilde{\Phi}(\mathbf{x} + \mathbf{s}_\mu) - \Omega(\mathbf{x})\tilde{\Phi}(\mathbf{x})}{s_\mu} - i\Omega(\mathbf{x})\tilde{A}_\mu(\mathbf{x})\tilde{\Phi}(\mathbf{x}) - \left(\frac{\Omega(\mathbf{x} + \mathbf{s}_\mu) - \Omega(\mathbf{x})}{s_\mu}\right)\tilde{\Phi}(\mathbf{x}) \\ &\neq \Omega \left[\frac{\tilde{\Phi}(\mathbf{x} + \mathbf{s}_\mu) - \tilde{\Phi}(\mathbf{x})}{s_\mu} + i\tilde{A}_\mu(\mathbf{x})\tilde{\Phi}(\mathbf{x}) \right]. \end{aligned} \quad (1.9)$$

The reason why this fails is because the usual covariant derivative effects an infinitesimal parallel transport of the fields, but on the lattice we need to parallel transport the fields across finite distances. Instead, we can introduce lattice link variables as a dynamical element of the symmetry group for each link in our lattice, $U_\mu(\mathbf{x}) \in G$, which are responsible for parallel transporting the field from $\mathbf{x}$ to $\mathbf{x} + \mathbf{s}_\mu$. Using this, we can define covariant finite difference operators that preserve the gauge symmetry, even on the lattice, such as

$$\mathcal{D}_\mu\Phi \rightarrow \frac{\Phi(\mathbf{x} + \mathbf{s}_\mu) - U_\mu(\mathbf{x})\Phi(\mathbf{x})}{s_\mu}, \quad (1.10)$$

where $U_\mu(\mathbf{x})$ transforms as $U_\mu(\mathbf{x}) = \Omega(\mathbf{x} + \mathbf{s}_\mu)\tilde{U}_\mu(\mathbf{x})\Omega^\dagger(\mathbf{x})$, such that $\mathcal{D}_\mu\Phi = \Omega\tilde{\mathcal{D}}_\mu\tilde{\Phi}$. We can also act on this operator with $U_\mu^\dagger(\mathbf{x})$ to see that the hermitian conjugate of the lattice link variables perform parallel transport in the opposite direction — in this case from $\mathbf{x} + \mathbf{s}_\mu$ to $\mathbf{x}$. We can examine the continuum limit by reintroducing the gauge field as a generator of the lattice link variables, $U_\mu = e^{-is_\mu A_\mu}$, for which the Taylor expansion is $\mathbb{I} - is_\mu A_\mu + \mathcal{O}(s^2)$. Therefore, our covariant finite difference operator approximates the continuum covariant derivatives with errors that are of the order of the lattice spacings, just like in the global case.

The final piece that we need is a lattice version of the Yang-Mills term, $\text{Tr}[F_{\mu\nu}F^{\mu\nu}]$. Just like for the covariant derivatives, attempting to discretise the continuum expression for the field strength tensor, $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + i[A_\mu, A_\nu]$, breaks the gauge symmetry. We can construct a gauge invariant object by taking the trace of multiple parallel transport operations that return back to the starting point,

$$P_{\mu\nu}(\mathbf{x}) = \text{Tr}[U_\mu^\dagger(\mathbf{x})U_\nu^\dagger(\mathbf{x} + \mathbf{s}_\mu)U_\mu(\mathbf{x} + \mathbf{s}_\nu)U_\nu(\mathbf{x})]. \quad (1.11)$$

This is known as a Wilson loop and is often represented diagrammatically by the picture in Figure 1. Although we have not yet made it's connection to the Yang-Mills term explicit, it should not be surprising that they are related given that there is an obvious connection to the concept of holonomy. The gauge invariance of the Wilson loop can be seen by noting that the interior transforms as

$$U_\mu^\dagger(\mathbf{x})U_\nu^\dagger(\mathbf{x} + \mathbf{s}_\mu)U_\mu(\mathbf{x} + \mathbf{s}_\nu)U_\nu(\mathbf{x}) = \Omega(\mathbf{x})\tilde{U}_\mu^\dagger(\mathbf{x})\tilde{U}_\nu^\dagger(\mathbf{x} + \mathbf{s}_\mu)\tilde{U}_\mu(\mathbf{x} + \mathbf{s}_\nu)\tilde{U}_\nu(\mathbf{x})\Omega^\dagger(\mathbf{x}), \quad (1.12)$$

¹ It also requires $\Omega^\dagger\Omega$ because of the form of the derivative term that I have used. This will be the case for most symmetries that we are interested in, but there is potentially a generalisation to be made here.

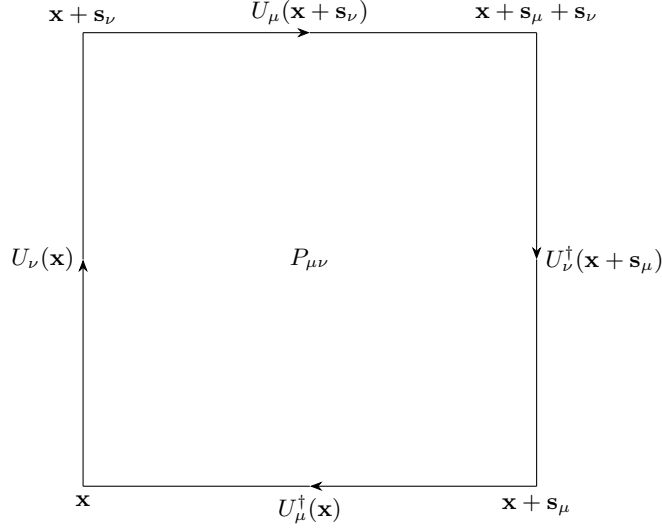


FIG. 1. A diagram of a Wilson loop.

which is just a similarity transformation. Since similar matrices share the same eigenvalues and the trace of a matrix is the sum of the eigenvalues, the Wilson loop is gauge invariant. Now let us make the connection to the Yang-Mills term explicit by expressing each lattice link variable in terms of the gauge field generators, as we did for the covariant derivatives, but now using the expansion $A_\mu(\mathbf{x} + \mathbf{s}_\nu) = A_\mu + s_\nu \partial_\nu A_\mu + \mathcal{O}(s^2)$,

$$\begin{aligned}
 P_{\mu\nu} &= \text{Tr} \left[e^{is_\mu A_\mu} e^{is_\nu (A_\nu + \mathbf{s}_\mu \partial_\mu A_\nu) + \mathcal{O}(s^3)} e^{-is_\mu (A_\mu + s_\nu \partial_\nu A_\mu) + \mathcal{O}(s^3)} e^{-is_\nu A_\nu} \right] \\
 &= \text{Tr} \left[\exp \left\{ i \left(s_\mu A_\mu + s_\nu A_\nu + s_\mu s_\nu \partial_\mu A_\nu + \frac{1}{2} i g s_\mu s_\nu [A_\mu, A_\nu] \right) + \mathcal{O}(s^3) \right\} \exp \left\{ -i \left(s_\mu A_\mu + s_\nu A_\nu \right. \right. \right. \\
 &\quad \left. \left. \left. + s_\mu s_\nu \partial_\nu A_\mu - \frac{1}{2} i s_\mu s_\nu [A_\mu, A_\nu] \right) + \mathcal{O}(s^3) \right\} \right] \\
 &= \text{Tr} \left[\exp \left\{ i s_\mu s_\nu (\partial_\mu A_\nu - \partial_\nu A_\mu + i [A_\mu, A_\nu]) + \mathcal{O}(s^3) \right\} \right] = \text{Tr} \left[e^{i s_\mu s_\nu F_{\mu\nu} + \mathcal{O}(s^3)} \right], \tag{1.13}
 \end{aligned}$$

where we have made use of the Baker–Campbell–Hausdorff formula, $e^A e^B = e^{A+B+\frac{1}{2}[A,B]+\dots}$ and it should be understood that all terms without an explicit dependence on position are assumed to be at $\mathbf{x}$. It is now clear how the Wilson loop relates to the field strength and the exponential can be expanded out into

$$P_{\mu\nu} = \text{Tr} \left[\mathbb{I} + i s_\mu s_\nu F_{\mu\nu} - \frac{1}{2} s_\mu^2 s_\nu^2 F_{\mu\nu} F_{\mu\nu} + \mathcal{O}(s^6) \right], \tag{1.14}$$

where there is no sum over the indices, they are simply labels of the directions. **[This is the typical derivation but it is slightly sloppy as there can be additional factors of order 3 and 4 appearing from the terms that I dropped at order 3. I think that odd powers in s all cancel, but I haven't shown that yet, and I'm not sure why the order 4 term does not contribute...]** Therefore, we can approximate the Yang-Mills term on the lattice by summing over all possible plaquettes, which each contribute to the Lagrangian through

$$\text{Tr}[F_{\mu\nu} F_{\mu\nu}] = \frac{1}{s_\mu^2 s_\nu^2} (2\text{Tr}[\mathbb{I}] - P_{\mu\nu} - P_{\mu\nu}^*) + \mathcal{O}(s^2), \tag{1.15}$$

where we have assumed that either $F_{\mu\nu}$ is real or that $\text{Tr}[F_{\mu\nu}] = 0$, which covers most typical scenarios such as U(1) or SU(N). For the purposes of simulations, we can choose to work in the temporal gauge, which sets $U_0(\mathbf{x}) = \mathbb{I}$. This is not a complete gauge fixing, as there is a residual freedom to make gauge transformations that do not depend upon

time. In this gauge, a typical action looks like (with overall factors of lattice spacings factored out)

$$\begin{aligned}
S \rightarrow & \sum_{\mathbf{x}, t} \left\{ \frac{1}{2} \left(\frac{\Phi^\dagger(\mathbf{x}, t + \Delta t) - \Phi^\dagger(\mathbf{x}, t)}{\Delta t} \right) \left(\frac{\Phi(\mathbf{x}, t + \Delta t) - \Phi(\mathbf{x}, t)}{\Delta t} \right) \right. \\
& - \frac{1}{2} \sum_i \left(\frac{\Phi^\dagger(\mathbf{x} + \mathbf{s}_i, t) - \Phi^\dagger(\mathbf{x}, t) U_i^\dagger(\mathbf{x}, t)}{s_i} \right) \left(\frac{\Phi(\mathbf{x} + \mathbf{s}_i, t) - U_i(\mathbf{x}, t) \Phi(\mathbf{x}, t)}{s_i} \right) \\
& \left. + \sum_i \frac{4}{g^2 s_i^2 \Delta t^2} \left(\text{Tr}[\mathbb{I}] - \text{Re}[P_{i0}(\mathbf{x}, t)] \right) - \sum_{i, j < i} \frac{4}{g^2 s_i^2 s_j^2} \left(\text{Tr}[\mathbb{I}] - \text{Re}[P_{ij}(\mathbf{x})] \right) - V(\Phi(\mathbf{x}, t)) \right\}, \quad (1.16)
\end{aligned}$$

where to keep it simple we are only considering Minkowski space with Cartesian coordinates. The equation of motion for the scalar field is determined by varying $\Phi^\dagger(\mathbf{x}, t) \rightarrow \Phi^\dagger(\mathbf{x}, t) + \delta\Phi^\dagger(\mathbf{x}, t)$ and requiring extremisation of the action,

$$\begin{aligned}
& - \left(\frac{\Phi(\mathbf{x}, t + \Delta t) - 2\Phi(\mathbf{x}, t) + \Phi(\mathbf{x}, t - \Delta t)}{\Delta t^2} \right) \\
& + \sum_i \left(\frac{U_i^\dagger(\mathbf{x}, t) \Phi(\mathbf{x} + \mathbf{s}_i, t) - 2\Phi(\mathbf{x}, t) + U_i(\mathbf{x} - \mathbf{s}_i, t) \Phi(\mathbf{x} - \mathbf{s}_i, t)}{s_i^2} \right) - \frac{\partial V}{\partial \Phi^\dagger} = 0. \quad (1.17)
\end{aligned}$$

The link variables must be members of the symmetry group, so the variation is restricted compared to the variation of the scalar field. We can vary them with $U_i(\mathbf{x}, t) \rightarrow (\mathbb{I} + i\epsilon^a T^a) U_i(\mathbf{x}, t)$, where ϵ^a are the perturbations and T^a are the generators of the group. Additionally, we need to take the variation $U_i^\dagger(\mathbf{x}, t) \rightarrow U_i^\dagger(\mathbf{x}, t) (\mathbb{I} - i\epsilon^a T^{a\dagger})$ into account because they are not independent variables. Under this variational procedure, we get the equations of motion

$$\begin{aligned}
& \text{Im} \left(- \frac{1}{\Delta t^2} \text{Tr} \left[U_i^\dagger(\mathbf{x}, t) T^{a\dagger} U_i(\mathbf{x}, t + \Delta t) \right] + \frac{1}{\Delta t^2} \text{Tr} \left[U_i^\dagger(\mathbf{x}, t - \Delta t) T^a U_i(\mathbf{x}, t) \right] \right. \\
& + \sum_{j \neq i} \frac{1}{s_j^2} \text{Tr} \left[U_i^\dagger(\mathbf{x}, t) T^{a\dagger} U_j^\dagger(\mathbf{x} + \mathbf{s}_i, t) U_i(\mathbf{x} + \mathbf{s}_j, t) U_j(\mathbf{x}, t) \right] \\
& - \sum_{j \neq i} \frac{1}{s_j^2} \text{Tr} \left[U_i^\dagger(\mathbf{x} - \mathbf{s}_j, t) U_j^\dagger(\mathbf{x} - \mathbf{s}_j + \mathbf{s}_i, t) T^a U_i(\mathbf{x}, t) U_j(\mathbf{x} - \mathbf{s}_j, t) \right] \\
& \left. + \frac{g^2}{4} \Phi^\dagger(\mathbf{x}, t) U_i^\dagger(\mathbf{x}, t) T^{a\dagger} \Phi(\mathbf{x} + \mathbf{s}_i, t) \right) = 0. \quad (1.18)
\end{aligned}$$

It will often turn out to be convenient to define $\bar{U}_i = U_i(\mathbf{x}, t + \Delta t) U_i^\dagger(\mathbf{x}, t)$ and update the link variables in two stages — firstly calculating $\bar{U}$ from the equation of motion and then by using $U_i(\mathbf{x}, t + \Delta t) = \bar{U} U_i(\mathbf{x}, t)$. Note that we can use the cyclic property of the trace to bring the generators to the front.

There is also a constraint equation that is associated with the gauge fixing that we have chosen. If I reintroduce the dependence on $U_0(\mathbf{x}, t)$ that I have omitted from the action, I can perform the variational procedure with this degree of freedom to find an additional equation of motion,

$$\begin{aligned}
& \text{Im} \left(\sum_i \frac{1}{s_i^2} \text{Tr} \left[U_i^\dagger(\mathbf{x}, t) U_i(\mathbf{x}, t + \Delta t) T^a \right] - \sum_i \frac{1}{s_i^2} \text{Tr} \left[U_i^\dagger(\mathbf{x} - \mathbf{s}_i, t) T^{a\dagger} U_i(\mathbf{x} - \mathbf{s}_i, t + \Delta t) \right] \right. \\
& \left. - \frac{g^2}{4} \Phi^\dagger(\mathbf{x}, t) T^{a\dagger} \Phi(\mathbf{x}, t + \Delta t) \right) = 0, \quad (1.19)
\end{aligned}$$

where I have reimposed that $U_0 = \mathbb{I}$ at the end of the process. This is a constraint equation that would correspond to Gauss's law in an electromagnetic context. Since it is not an evolution equation, it cannot be directly imposed during simulations, instead we must choose an initial configuration that respects this gauge fixing condition and check that it remains true throughout the evolution. A large part of the motivation for a lattice gauge theory approach to simulations is that a naive discretisation of continuum equations of motion for the gauge fields will result in an evolution with rapidly growing violation of this condition. **It would be good to have a concrete understanding of why this is prevented in the lattice gauge approach. I think there is something like a discrete version of a conserved current (because we have a discrete version of gauge invariance), which ensures that the violation of this condition is only produced by the limitations of machine precision.**

II. SPECIFIC MODELS

When it comes to simulating gauged systems on a lattice, it is usually preferable to evolve the generators of the lattice link variables, because there is typically a smaller number of fields to evolve than all of the components of the matrix-valued link variables themselves, and it guarantees that a given link variable remains a member of the symmetry group. The specific approach will depend upon the exact model being simulated and we will describe some models of interest below.

A. The Standard Model

The Lagrangian for the Higgs sector of the Standard Model is

$$\mathcal{L} = \frac{1}{2}(\mathcal{D}_\mu \Phi)^\dagger (\mathcal{D}^\mu \Phi) - \frac{1}{g^2} \text{Tr}[W_{\mu\nu} W^{\mu\nu}] - \frac{1}{g'^2} Y_{\mu\nu} Y^{\mu\nu} - V(\Phi), \quad (2.1)$$

where the continuum version of the covariant derivative is

$$\mathcal{D}_\mu = \partial_\mu + iW_\mu + iY_\mu \quad (2.2)$$

and the field strength tensors are

$$W_{\mu\nu} = \partial_\mu W_\nu - \partial_\nu W_\mu + i[W_\mu, W_\nu], \quad Y_{\mu\nu} = \partial_\mu Y_\nu - \partial_\nu Y_\mu, \quad (2.3)$$

which can all be related to the analysis in the previous section by the appropriate substitution for A_μ . If I define $W_\mu = \frac{g}{2} \mathcal{W}_\mu^a \sigma^a$ and $Y_\mu = \frac{g'}{2} \mathcal{Y}_\mu$, where σ^a are the Pauli matrices, then the Lagrangian definition is in the form that I am used to, with the gauge couplings appearing inside the derivative and field strength tensor in the case of SU(2).

In principle we could treat the lattice link variables as elements of U(2), but since they have different gauge couplings it is more convenient to break them down into $U_{W\mu} = e^{-i w_\mu^a \sigma^a}$ and $U_{Y\mu} = e^{-i y_\mu}$, with $w_\mu^a = \frac{1}{2} g s_\mu W_\mu^a$ and $y_\mu = \frac{1}{2} s_\mu Y_\mu$. Since we will frequently need to evaluate the lattice link variables, it is not sufficient to define them in terms of the expansion of the exponentials. Instead we will use the equivalent definitions

$$U_{W\mu} = \cos(\bar{w}_\mu) \sigma^0 - i \sin(\bar{w}_\mu) \hat{w}_\mu^a \sigma^a, \quad U_{Y\mu} = \cos(y_\mu) - i \sin(y_\mu), \quad (2.4)$$

where σ^0 is the 2×2 identity matrix and we have defined $w_\mu^a = \bar{w}_\mu \hat{w}_\mu^a$ with $\hat{}$ denoting a unit vector. There are other options for encoding these matrices, particularly in SU(2) which it is worth briefly mentioning. This method encodes the matrices with close to the minimum required amount of memory — it is possible to reduce it further using choices of gauge, such as the temporal gauge discussed above but this is not a complete gauge fixing so there is some potential for improvement. However, this comes at the cost of having to perform additional computations to “undo” the encoding and get the link variables in a convenient form for further calculations. An alternative option, which has been implemented in the code, is to use the quaternionic representation of SU(2),

$$U_{W\mu} = c_\mu^0 \sigma^0 + i c_\mu^a \sigma^a, \quad (2.5)$$

with the constraint that $c^0 c^0 + c^a c^a = 1$, which is clearly closely related to the previous expression. This uses more memory, simply because 4 variables are stored for each element of SU(2) rather than 3, but has the advantage that it is already in a convenient form for further calculations. There is, therefore, a trade-off between memory usage and speed (probably the quaternionic representation also has slightly less numerical error) depending upon how the SU(2) matrices are represented numerically.

The lattice version of the covariant derivative in the SM is

$$\mathcal{D}_\mu \Phi \rightarrow \frac{\Phi(\mathbf{x} + \mathbf{s}_\mu) - U_{Y\mu}(\mathbf{x}) U_{W\mu}(\mathbf{x}) \Phi(\mathbf{x})}{s^\mu}. \quad (2.6)$$

The equations of motion are very similar to the case above; the scalar field equation is exactly the same with $U_i = U_{Yi} U_{Wi} \in \text{U}(2)$, but here we have two separate Yang-Mills terms with different gauge couplings, and therefore there are two copies of equation (1.18), that interact through the current term. For the hypercharge equation, there are a lot of simplifications to be made because of the simplicity of the group. There is only one generator, which is

the identity, and the traces can be ignored. By using the expression above, we arrive at,

$$\begin{aligned} & \text{Im} \left(-\frac{1}{\Delta t^2} \text{Tr} \left[U_i^\dagger(\mathbf{x}, t) T^{a\dagger} U_i(\mathbf{x}, t + \Delta t) \right] + \sum_{j \neq i} \frac{1}{s_j^2} \text{Tr} \left[U_i^\dagger(\mathbf{x}, t) T^{a\dagger} U_j^\dagger(\mathbf{x} + \mathbf{s}_i, t) U_i(\mathbf{x} + \mathbf{s}_j, t) U_j(\mathbf{x}, t) \right] \right. \\ & \left. - \frac{g^2}{4} \Phi^\dagger(\mathbf{x}, t) U_i^\dagger(\mathbf{x}, t) T^{a\dagger} \Phi(\mathbf{x} + \mathbf{s}_i, t) \right) = 0. \end{aligned} \quad (2.7)$$